Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Need for statistics: purposes, and the collection and tabulation of data. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

Lesson and revision-note context

Subject	Mathematics
Grade	JSS1
Term	Third Term
Week	9
Topic	Need for statistic
Subtopic	Need for statistics: purposes, and the collection and tabulation of data
Theme	—
Curriculum objectives	• explain the purpose of statistic • analyse and interpret statistical data presented in tables
Previous knowledge	• Students are familiar with counting and recording information.
Lesson sequence	• explain the purpose of statistic — the teacher introduces the concept of statistics as a method of collecting and organising information • analyse and interpret statistical data presented in tables — the teacher takes the students through collecting data, recording it with tally marks, and setting it out in a frequency table
Instructional materials	• classroom register, bottle tops, pencils or counters.
Evaluation	• (a) Define statistic • (b) State three methods of collecting data. • (c) Prepare a frequency table for the following data: 1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2.

Lesson objectives

1.	explain the purpose of statistic
2.	analyse and interpret statistical data presented in tables

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

STUDENT REVISION NOTE

Topic: Need for statistics: purposes, and the collection and tabulation of data

Class: JSS1 **Subject:** Mathematics **Week:** 9

Section 1: What Statistics Is and What It Is For

When you count the pencils in your pencil case, or the number of learners in your class, you are dealing with information. That information — the facts or numbers you collect — is called data. Data are facts or information collected for a particular purpose.

Statistics is the science of collecting, organising, presenting, analysing and interpreting data. Let us look at each of those words:

- Collecting means gathering the facts or information you need.
- Organising means putting that information into some order, so it is easy to read.
- Presenting means showing the information clearly, for example in a table.
- Analysing means studying it carefully to see what it tells you.
- Interpreting means explaining what the information means.

So statistics is the whole process of working with data — from collecting it, right through to explaining what it shows.

Why do we bother with all of this? Statistics is important because it helps us to:

- Collect information — gather the facts we need in an orderly way.
- Organise information — arrange the facts so they are easy to manage and read.
- Compare information — put facts side by side and see how they are alike or different.
- Make decisions — use the facts to choose sensibly, rather than guessing.
- Present information clearly — show the facts so that other people can understand them at a glance.

In short, statistics gives us a reliable way to handle information, so that the choices we make are based on real facts.

Think about the facts of your own class. The number of boys and the number of girls are pieces of data — facts collected about your class. When you write them in a table so they are easy to read, you are organising the data. When you look at the two numbers and say which group is larger, you are comparing the information. And if you use that comparison to decide, say, how many chairs the class needs, you have used statistics to make a decision. One set of data can pass through every part of statistics in just a few steps.

Practice

1. What is statistics?
2. What are data?
3. Name three ways in which statistics is important.
4. A shop keeper writes down how many bottles of drink she sells each day of the week so she can see which day she sells the most. Which two uses of statistics is she showing?

Section 2: Collecting Data with Tally Marks and Building a Frequency Table

When we want to answer a question like "How many cars pass our school each morning?" we first have to gather the information. That gathered information is called **data**. Data can be collected in four main ways:

1. **Observation** — you watch and note down what you see.
2. **Counting** — you count things, such as the number of students present.
3. **Interview** — you ask people questions and record their answers.

4. **Questionnaire** — you hand out a set of written questions for people to answer.

Once the data is collected, it can be a jumbled list, which is hard to read. So we tidy it up using **tally marks** and a **frequency table**.

A **tally mark** is a single stroke (|) used to record one item. To make counting easy, every fifth item is marked by drawing a stroke across the previous four, like this: ||||. A group of five tally marks is always drawn as four vertical strokes with one stroke across them. Each group stands for 5.

A **frequency table** is a table showing how many times each item occurs in a set of data. The number of times an item appears is called its **frequency**.

To build a frequency table, you go through your data one item at a time. Each time you meet an item, add a tally stroke next to it. When an item reaches five strokes, you close the group by drawing the fifth stroke across. At the end, count the strokes in each row and write that number as the total (frequency).

Worked example — recording vehicles with tally marks

Two students collected the same data about the vehicles that passed their school:

bus, car, car, car, lorry, bicycle, bicycle, car, car, lorry, bicycle, car

This jumbled list is hard to read, so let us record it with tally marks and show it in a table.

Work through the list one vehicle at a time and add a stroke each time:

- **car** appears 6 times. Its tally is one group of five plus one more: |||| |.
- **bus** appears 1 time. Its tally is |.
- **lorry** appears 2 times. Its tally is ||.
- **bicycle** appears 3 times. Its tally is |||.
- **motor bike** appears 0 times, so it gets no tally marks.

Now we count the strokes in each row to find the total:

Vehicles	Tally	Total
car		
bus	|	1
lorry	||	2
bicycle	|||	3
motor bike		0

This table shows at a glance which vehicle was most common (car, with 6) and which was least common (motor bike, with 0). The "Total" column is the frequency of each vehicle.

Worked example — a frequency table of favourite colours

A class of students was asked for their favourite colour. The results were counted and written as a frequency table:

colour	frequency
Red	18
Blue	20
Green	14
Yellow	16
White	15
Pink	30

This table shows how many students chose each colour. The frequency of a colour is just the number of students who chose it. For example, the frequency of **Pink** is 30, which means 30 students said pink was their favourite. Reading the table, we can compare the colours at a glance: Blue (20) and Pink (30) were chosen more than Green (14).

Practice

1. Prepare a frequency table for the following data:

   1, 1, 3, 4, 4, 3, 2, 2, 3, 3, 4, 1, 2

   Set out a table with columns for the value, its tally, and its total (frequency).
2. The tally row below records how many times a certain colour appeared in a set of data. What is the frequency (total) of that colour?

   |||| ||
3. Name three methods of collecting data.

Check your work

Attempt every question before you read this page.

What Statistics Is and What It Is For

1. Statistics is the science of collecting, organising, presenting, analysing and interpreting data.
2. Data are facts or information collected for a particular purpose.
3. Any three of: it helps us to collect information, organise information, compare information, make decisions, and present information clearly.
4. She is collecting information (writing down the daily sales) and comparing information (seeing which day has the most sales) to make a decision.

Collecting Data with Tally Marks and Building a Frequency Table

1. Value 1: 3; Value 2: 3; Value 3: 4; Value 4: 3.

   Go through the data and count how many times each value occurs:
   - 1 appears in positions 1, 2 and 12 → 3 times
   - 2 appears in positions 7, 8 and 13 → 3 times
   - 3 appears in positions 3, 6, 9 and 10 → 4 times
   - 4 appears in positions 4, 5 and 11 → 3 times

   The frequency table is:

Value	Tally	Total
1		
2		
3						4
4		
2. 7

   The first group |||| is a completed group of 5, drawn as four vertical strokes with one across. The two strokes after it are two more items. So the total is $5 + 2 = 7$.
3. Any three of: observation, counting, interview, questionnaire.

Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

Revision note ID	cand-c5e2aa11	**Date**	
Subtopic	Need for statistics: purposes, and the collection and tabulation of data	**Rater**	

> The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

Criterion (select one score per row)	**1** Poor	**2** Weak	**3** Adequate	**4** Good	**5** Excellent
Objective and curriculum alignment Does the revision note cover the intended lesson objectives and curriculum target?	○	○	○	○	○
Taught-lesson alignment Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth?	○	○	○	○	○
Scope and conceptual correctness Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors?	○	○	○	○	○
Student-facing clarity and usefulness Can JSS1 learners understand and use it meaningfully as take-home review material?	○	○	○	○	○

Strengths	Concerns	Recommendation

Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

☐	**Misses a required objective** — Skips something the lesson was meant to teach.
☐	**Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic).
☐	**Wrong or misleading content** — Contains a mathematical or factual mistake.
☐	**Too hard for JSS1 to follow** — Likely confusing for the students.
☐	**Other:** _____